historyless variants of GPT-4o achieve the lowest average formation energies, with −1.97 eV/atom and −1.99 eV/atom, respectively. GPT-4o with history also achieves the lowest minimum formation energy per atom at −2.72 eV/atom. Interestingly, while the minimum formation energy per atom values achieved by Gemini-1.0-pro are close to that of GPT-4o, its average formation energy per atom values are significantly higher, indicating that it struggles to consistently suggest chemically stable modifications for the materials. Nonetheless, Gemini-1.0-pro still noticeably outperforms the baseline. In Fig. D.1 and D.2, we visualize 20 materials discovered by LLMatDesign for the band gap and formation energy tasks, respectively. These materials are obtained from the first run of all 10 starting materials. For the band gap task, the final materials are selected. For the formation energy task, the materials with the lowest formation energy per atom are chosen.

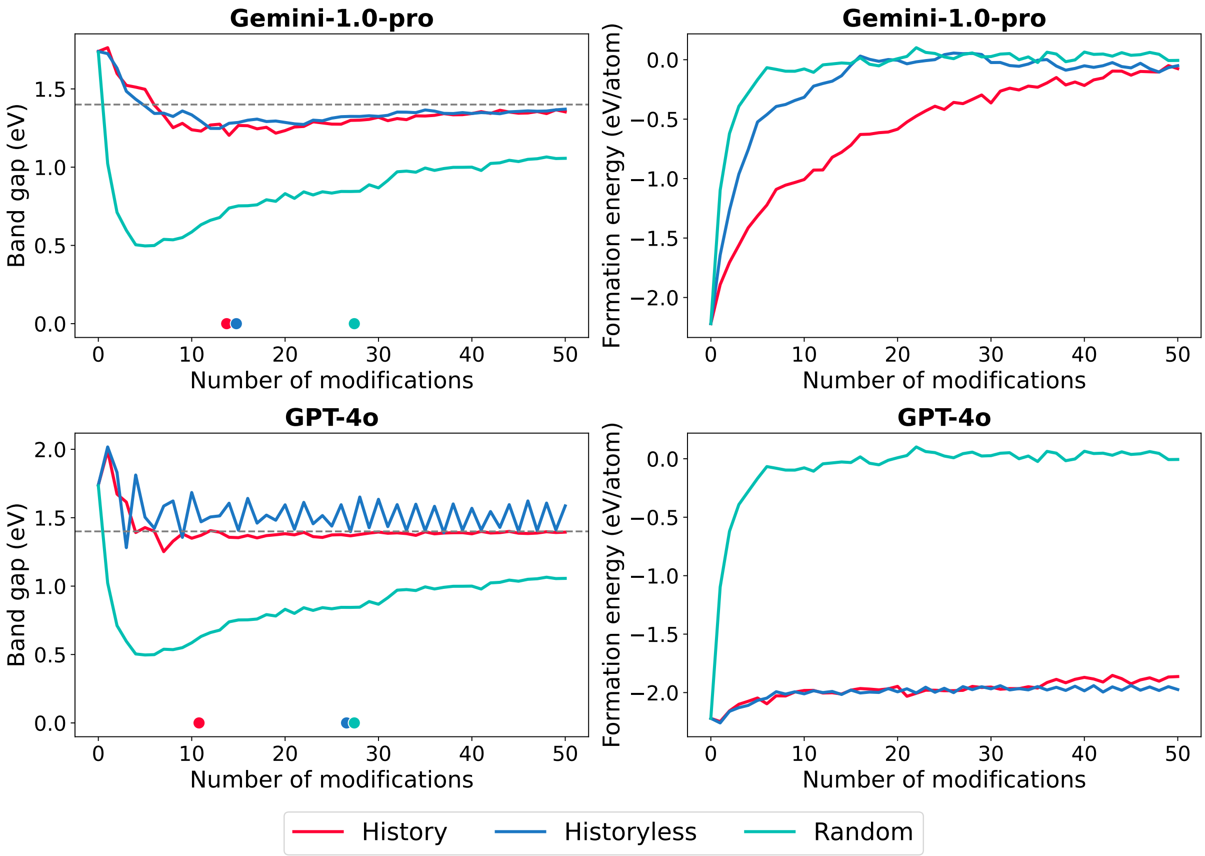
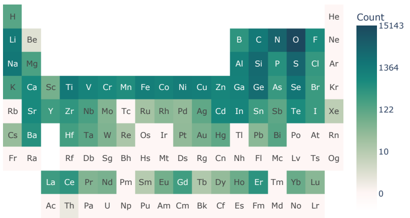


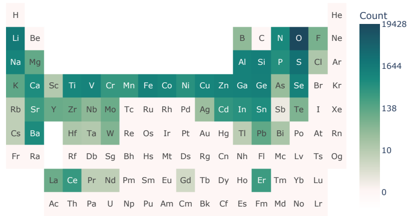
Figure 4: Average band gaps and formation energies over 50 modifications. The grey horizontal line indicates the target band gap of 1.4 eV. The colored dots on the x-axis indicate the average number of modifications taken for each method to reach the target. For formation energy, the goal is to achieve the lowest possible value.

In Fig. 4, we plot the band gaps and formation energies per atom over 50 modifications, averaged across 10 starting materials. The target band gap of 1.4 eV is indicated by the grey horizontal line. Both history and historyless variants of Gemini-1.0-pro and GPT-4o demonstrate quick convergence to the target band gap. However, the GPT-4o historyless variant exhibits zig-zag oscillations in band gap values as modifications increase. This occurs because, without historical information, GPT-4o tends to oscillate between a few of the same moves, causing the band gap to fluctuate without improving. In contrast, the random baseline fails to converge to 1.4 eV within the maximum allowed 50 modifications. For formation energy, our findings indicate that GPT-4o is consistently able to suggest modifications which keep formation energy low on average around −2 eV/atom, though Gemini-1.0-pro struggles to do so despite being able to obtain a low minimum formation energy. Notably, neither GPT-4o nor Gemini-1.0-pro are able to beat the formation energy of the starting materials, likely due to the the fact that these materials are already at or near the lowest energy states.

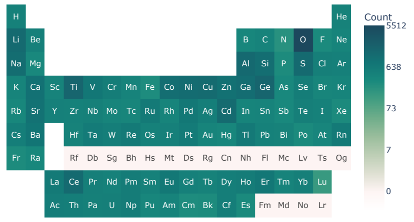
Fig. 5 presents heatmaps over the periodic table displaying the element occurrences in the modifications for both the band gap and formation energy tasks, which reveal additional insights into the reason for the good performance for LLM-driven design. The number of occurrences of each element is collected across all runs and starting materials. In the heatmaps for the random baseline, all elements are chosen at nearly uniform frequencies. This result is to be expected, as the random algorithm samples elements with atomic numbers up to 99 uniformly. Meanwhile, in the heatmaps for the LLM cases, there is a clear distribution towards certain elements, mostly focusing on elements within the first four rows of the periodic table and avoiding noble



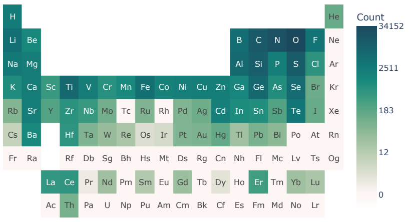
BG: Gemini-1.0-pro with history



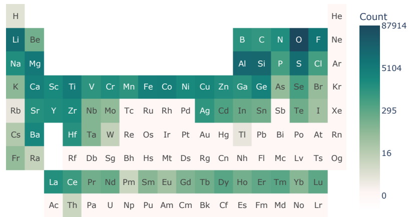
BG: GPT-4o with history



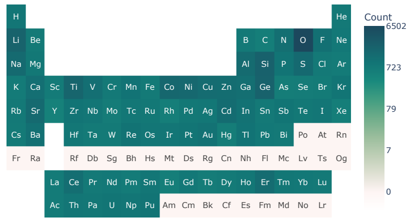
BG: Random



FE: Gemini-1.0-pro with history



FE: GPT-4o with history



FE: Random

Figure 5: Heatmaps of element frequencies in band gap (BG) and formation energy (FE) tasks. The periodic table is color-coded to indicate the frequency of each element's occurrence in all modified materials (both intermediate and final) across all runs and starting materials. Darker colors represent higher frequencies, while lighter colors denote lower frequencies or absence. The visualization employs log-scaling to effectively highlight the distribution and prevalence of elements.

metals and Actinides. Both LLM models share similar distributions, such as a preference for elements like oxygen, however Gemini-1.0-pro's suggestions appear to exhibit a greater element diversity compared to GPT-4o, including some of the transition metals. With Gemini-1.0-pro, we also occasionally observe modifications suggested by the LLM that include noble gases, which is not chemically feasible due to their inert nature. With GPT-4o, this does not occur (see Fig. C.1). Regardless, both LLM models are able to consistently suggest chemically viable elements for modification, which is akin to how a human expert would make similar choices based on chemical intuition or from past examples in the literature.

In Fig. 6, we present an example of the full process whereby LLMatDesign successfully completes a design task to achieve a band gap of 1.40 eV. In the first step, LLMatDesign suggests modifying the starting material $CdCu_2GeS_4$ by substituting S with Se, given the hypothesis that increasing atomic radius and changing the electronegativity can alter the band gap. Upon modification, the new material $CdCu_2GeSe_4$ was found to have an even smaller band gap, which is contrary to the desired effect as noted by the reflection. This history is included in the second step of modification, whereby LLMatDesign suggests a subsequent modification of Ge with Si, which increases the gap. The reflection notes a partial success is achieved, but is still not enough to reach the target, whereupon a third step is taken. In the third step, Cu is substituted with Zn, which finally achieves the desired band gap within an acceptable threshold, ending the process. From this example, we can observe the LLM is successful at 1) recognizing differences in element properties (i.e. Se having a larger atomic radius than S), 2) highlighting these properties as being relevant to the design task (i.e. atomic radius, electronegativity, and electronic configuration affecting the band hap), 3) and recognizing whether a modification is successful and the degree of success in the reflection. We will show in the subsequent section that it is this reasoning and reflection process which has a significant impact on its success.

In the final step of the design process, a DFT calculation is performed to validate the material's properties which were obtained from a ML surrogate model. Here, we use DFT to compute the formation energy of the minimum energy structures in all 30 runs for each 10 starting materials obtained with GPT-4o and random sampling. On average, structures generated by LLMatDesign using GPT-4o with history achieved a formation energy of -2.32 eV/atom with a job completion rate of 73.3%. In comparison, the random baseline obtained an average formation energy of -1.51 eV/atom, with a significantly lower job completion rate of 40.0% (see Fig. E.1). The lower completion rate is likely due to a much higher likelihood of unreasonable compositions and configurations obtained from random sampling which cause the DFT calculations to fail. These results also follow the same relative trends in Table 2. In a situation where DFT is directly used in each iteration rather than a surrogate model, this step can be omitted, or potentially the materials can be validated further with experiments.

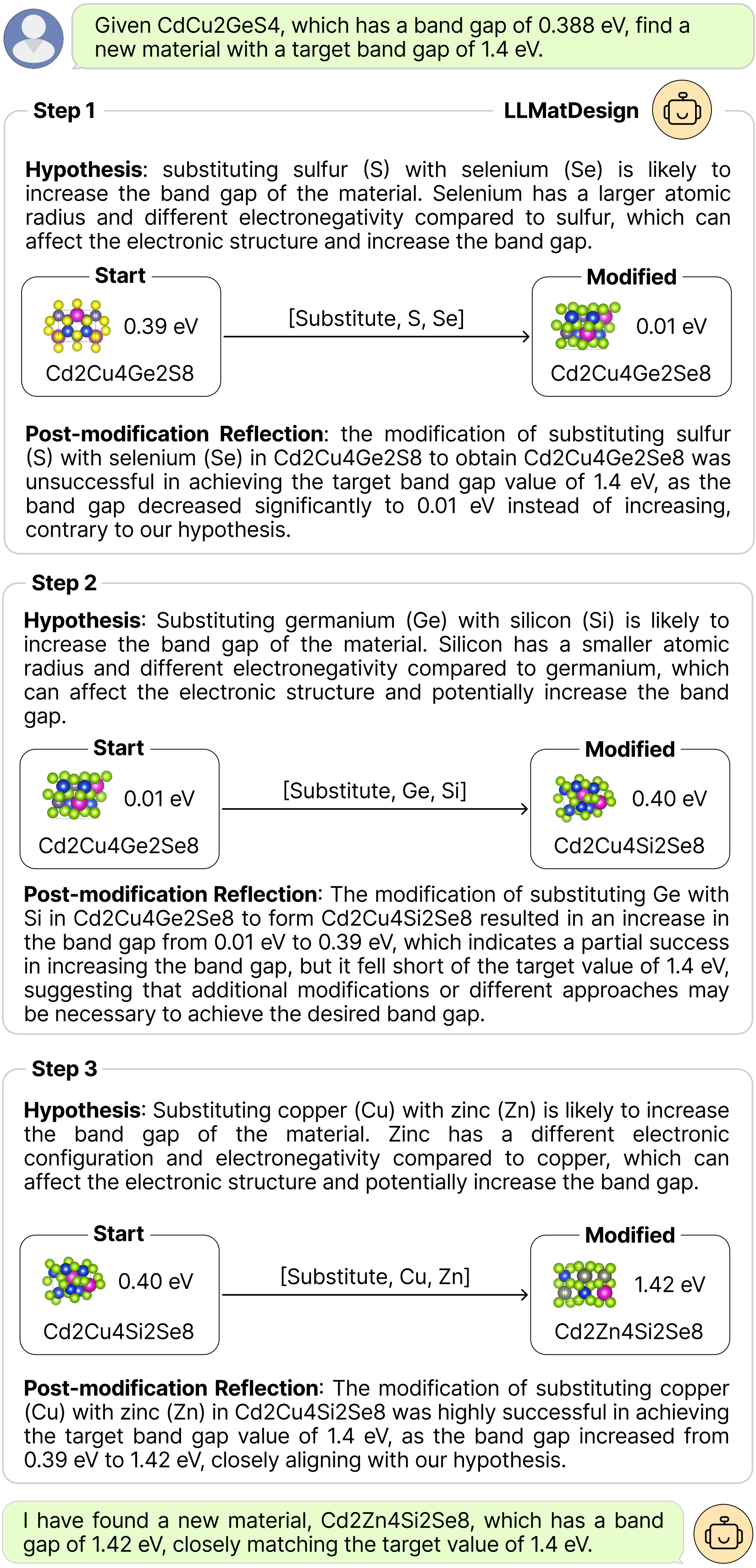


Figure 6: Example of LLMatDesign with GPT-4o on the task of modifying the starting material $CdCu_2GeS_4$ to achieve a band gap of 1.40 eV. The starting material is retrieved from the Materials Project with chemical formula $Cd_2Cu_4Ge_2S_8$.

2.3 Self-reflection

To quantify the effect of self-reflection on the performance of LLMatDesign, we conduct band gap experiments using GPT-4o and the same set of 10 starting materials, where we aim to find a new material with a target band gap of 1.4 eV. Like with the history variant, past modifications are incorporated into the prompting loop. However, in this case, self-reflection is omitted completely. In other words, the history message only includes the modification and hypothesis pairs (see Algo. 1). The results from these experiments are shown in Table 3. As previously discussed, GPT-4o with history achieves an average of 10.8 modifications, while GPT-4o without history requires 26.6 modifications. In comparison, GPT-4o with history but without self-reflection now needs an average of 23.4 modifications, which is over twice as many compared to including self-reflection. These results suggest that self-reflection, which involves the LLM evaluating and reasoning through its previous design choices, plays a crucial role in enhancing the efficiency of LLMatDesign in achieving the given objective.

Table 3: LLMatDesign with and without self-reflection. GPT-4o is used as the LLM engine.

Starting Material	Average # of Modifications		
	History	Historyless	History without reflection
$BaV_2Ni_2O_8$	17.7	30.4	45.1
$CdCu_2GeS_4$	3.3	9.5	5.0
$CeAlO_3$	7.4	16.9	27.6
Co_2TiO_4	5.5	1.6	7.9
$ErNi_2Ge_2$	19.3	47.6	31.0
Ga_2O_3	12.7	37.7	13.1
Li_2CaSiO_4	14.3	29.3	31.4
LiSiNO	4.1	2.8	5.1
Na_2ZnGeO_4	11.5	49.4	31.5
$SrTiO_3$	12.0	40.6	36.7
Avg.	**10.8**	26.6	23.4

2.4 Prompting

Well-crafted prompts are essential for eliciting accurate and useful responses from LLMs. While the base prompt template, shown in Fig. 2, works as intended, we subsequently show that optimizing this prompt can improve the performance of LLMatDesign even further. To this end, we develop two additional prompt templates in a non-exhaustive demonstration. The first template, termed *GPT-4o Refined*, is an enhancement of the original prompt (Fig. 2) created by GPT-4o itself. This refinement includes rephrasing and reformatting parts of the original prompt and appending the following sentence: "Take a deep breath and work on this problem step-by-step. Your thoughtful and detailed analysis is highly appreciated." The second template, named *Persona*, mirrors the original prompt but incorporates the persona of a materials specialist. Specifically, it begins with a declaration that the LLM is a materials design expert working on developing new materials with specific properties. Detailed descriptions of these prompt templates are provided in Appendix A.

We conduct the same experiments on the band gap task using GPT-4o as the LLM engine for LLMatDesign across all 10 starting materials. The results, shown in Table 4, indicate that both the GPT-4o Refined and Persona prompt templates outperform the GPT-4o with history, with the GPT-4o Refined template achieving the best performance, requiring an average of only 8.69 modifications to complete the task. The improvement over the original prompt template indicates that careful prompt optimization can positively enhance the efficiency and accuracy of LLM-directed materials discovery frameworks, and that this process can even be performed by the LLM itself. This is a particularly intriguing discovery as it hints towards an unprecedented level of autonomy which can be enabled by LLMs, whereby the prompts and instructions in the framework can be continuously tuned in an automated manner with minimal human intervention.

Table 4: LLMatDesign with different prompts. GPT-4o is used as the LLM engine.

Starting Material	Average # of Modifications		
	History	GPT-4o Refined	Persona
$BaV_2Ni_2O_8$	17.7	13.4	9.6
$CdCu_2GeS_4$	3.3	3.1	5.3
$CeAlO_3$	7.4	7.2	8.7
Co_2TiO_4	5.5	8.9	11.9
$ErNi_2Ge_2$	19.3	11.9	11.7
Ga_2O_3	12.7	8.4	8.3
Li_2CaSiO_4	14.3	11.6	11.9
LiSiNO	4.1	5.6	1.0
Na_2ZnGeO_4	11.5	6.9	8.8
$SrTiO_3$	12.0	9.9	13.9
Avg.	10.8	**8.69**	9.11

2.5 Constrained Materials Design

Materials discovery with constraints ensures scientific, economic, and political viability. For instance, avoiding the use of rare earth metals can reduce dependency on limited and expensive resources, mitigate supply chain risks, and align with environmental and ethical standards. To this end, we evaluate LLMatDesign under three constraints limiting its action space. Experiments are conducted on the band gap task using the starting material $SrTiO_3$ with GPT-4o to test whether these constraints are obeyed. Like before, each experiment is repeated 30 times, and the percentage of modifications adhering strictly to the constraints is calculated across all runs. As shown in Table 5, LLMatDesign perfectly adheres to the constraints of "do not use Ba or Ca" and "do not modify Sr," achieving 100% compliant modifications. For the constraint "do not have more than 4 distinct elements," only 4 out of 509 modifications by LLMatDesign include 5 distinct elements, resulting in a high compliance rate of 99.02%. These results demonstrate LLMatDesign's robust capability in adhering to predefined constraints as described by natural language, an advantage unique to LLM-driven design.

Table 5: LLMatDesign with different constraints on $SrTiO_3$.

Constraint	% compliant modifications
Do not use Ba or Ca	100
Do not modify Sr	100
Do not have more than 4 distinct elements	99.02

2.6 Further Discussion

Through extensive experiments, we find LLMatDesign consistently outperforms baselines by a significant margin, demonstrating the viability of using LLM-based autonomous agents for materials discovery tasks under a limited budget. While the random baseline uniformly samples from a set of elements for modification (see Fig. 5), LLMatDesign, whether utilizing GPT-4o or Gemini-1.0-pro, exhibits inherent chemical knowledge, enabling it to provide chemically meaningful suggestions. Furthermore, GPT-4o accurately recognizes periodic trends such as atomic radius and electronegativity in its hypotheses and self-reflections in guiding its decisions. In contrast, Gemini-1.0-pro is more prone to errors in this regard, likely due to it being a less robust LLM. Further experiments also show the critical role of self-reflection in the performance of the LLM. This indicates that by reviewing and learning from its previous decisions, LLMatDesign can refine its future suggestions more effectively. This iterative learning process helps the model understand the implications of its modifications better, leading to quicker convergence. In general, it is evident that there are more complex underpinnings behind the remarkable effectiveness of LLM-driven design than simply predicting most likely outcomes.

This work also demonstrates the *lower-bound* capabilities of LLM-based design, which is performed without further fine-tuning in a zero-shot manner. A natural extension of this approach would be to further train LLMs